

Experiment No.-02

**Study of Ductile-Brittle-Transition Behaviors of
Metallic Alloys.**

Submitted by,

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Department: MME

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Objective: Construction of ductile-brittle transition curves of alloys with BCC, and FCC structures. Comparison of DBT curves.

Introduction:

The quest for materials that seamlessly blend strength with resilience pushes us to probe the intricacies of their behavior under diverse conditions. Among these intricacies, the ductile-brittle transition (DBT) stands as a captivating crossroads, where a material's capacity to absorb energy through ductile deformation abruptly yields to catastrophic brittle fracture. Understanding the factors governing this transition, particularly its dependence on temperature and crystal structure, unlocks a treasure trove of informed material selection and design possibilities.

This report explores the interesting topic of DBT curves, specifically focusing on comparing these curves for alloys with Body-Centered Cubic (BCC) and Face-Centered Cubic (FCC) crystal structures. We carefully examine the curves of different materials to show how the crystal structure strongly affects DBT behavior, revealing important insights for material engineering.

For BCC: BCC crystal structure has limited slip systems that restrict the movement of dislocations under stress. Consequently, BCC alloys tend to exhibit sharper DBT curves, indicating a narrow temperature range over which their behavior shifts from ductile to brittle. This heightened susceptibility to cleavage fracture at lower temperatures stems from the inherent difficulty of accommodating stress within the BCC lattice.

For FCC: FCC alloys boast broader DBT curves, signifying a gradual transition and significantly higher resilience at lower temperatures. This enhanced ductility arises from the inherent flexibility of the FCC lattice, its ability to readily accommodate stress through dislocation movement before reaching the critical threshold for crack initiation.

DBT curves tell a story about how materials behave under different temperatures and stress. Each point on the curve shows the temperature at which a material resists fracturing. A steeper curve means a sharper shift from ductile to brittle behavior, while a gentler slope indicates a smoother transition.

Comparing DBT curves of BCC and FCC alloys helps us understand their differences. We see how their crystal structures affect deformation, crack initiation, and overall behavior. This knowledge helps us design materials for specific uses, ensuring they perform well and stay safe in their intended temperature range. (Dieter, 1928)

Tools and Equipment:

- Specimen (Mild steel)
- Charpy pendulum impact test machine
- Emery paper (150, 320, 600, 800, 1200)
- Wet polishing machine
- Microscope

Procedure:

1) We used mild steel as our specimen and conducted the Charpy test in the material testing lab. The Charpy test was performed using a pendulum-type impact testing machine, and the specifications of the test setup are given below:

- A pendulum with a mass of 18.748 kg, a length of 825 mm, and an angle of swing of 160° .
- Specimen holder (for Charpy test).
- Striking edge (for Charpy tests).
- Lock lever and pendulum releaser.
- Pendulum brake.
- A calibrated dial with red and black indicators to measure the impact energy.
- Slide calipers and scale.

We checked the specimens (-75, -20, 14, and 100) for their standard dimensions using proper measurement techniques. For the Charpy test, the corresponding striking edge was fixed to the hammer (ball pin hammer from 140° for the Charpy test).

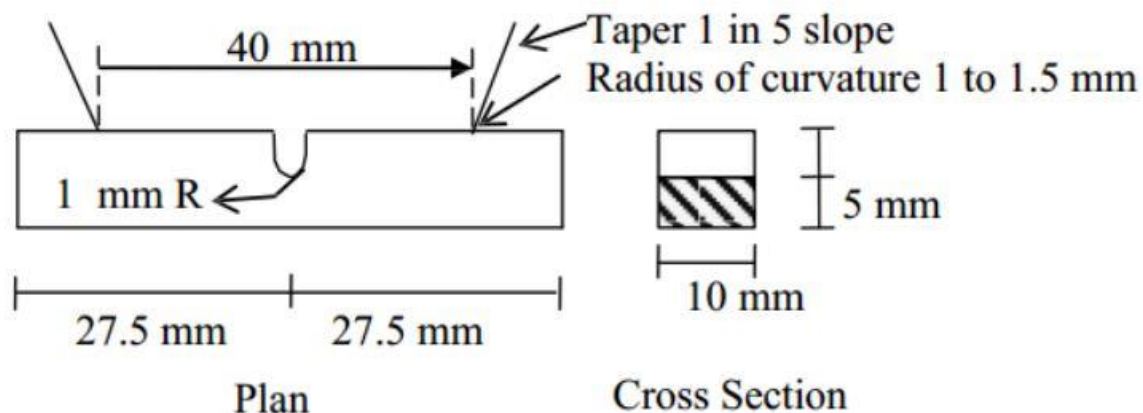


Figure 1: Standard specimen for Charpy test

3) The specimen was placed horizontally as a simple beam with the U-notch on the tension side.

4) The pendulum was raised to its highest position and locked. After releasing the pendulum by pressing down the lock lever, it struck the specimen. The damper plate/brake was used to stop the oscillations of the pendulum.

5) We recorded the dial readings for our 4 specimens.

6) Our specimen did not completely break. We also recorded the energy for our 4 samples at different temperatures.

7) After that, we took a part from our specimens (-75°C and 14°C) and conducted their microstructure analysis.

Data and Microstructure:

Cross section: 10*10mm, dia = 55mm

Material type and test temperature(°C) (Mild steel)	Impact Energy(J)
-75	242
-20	247
14	200
100	225

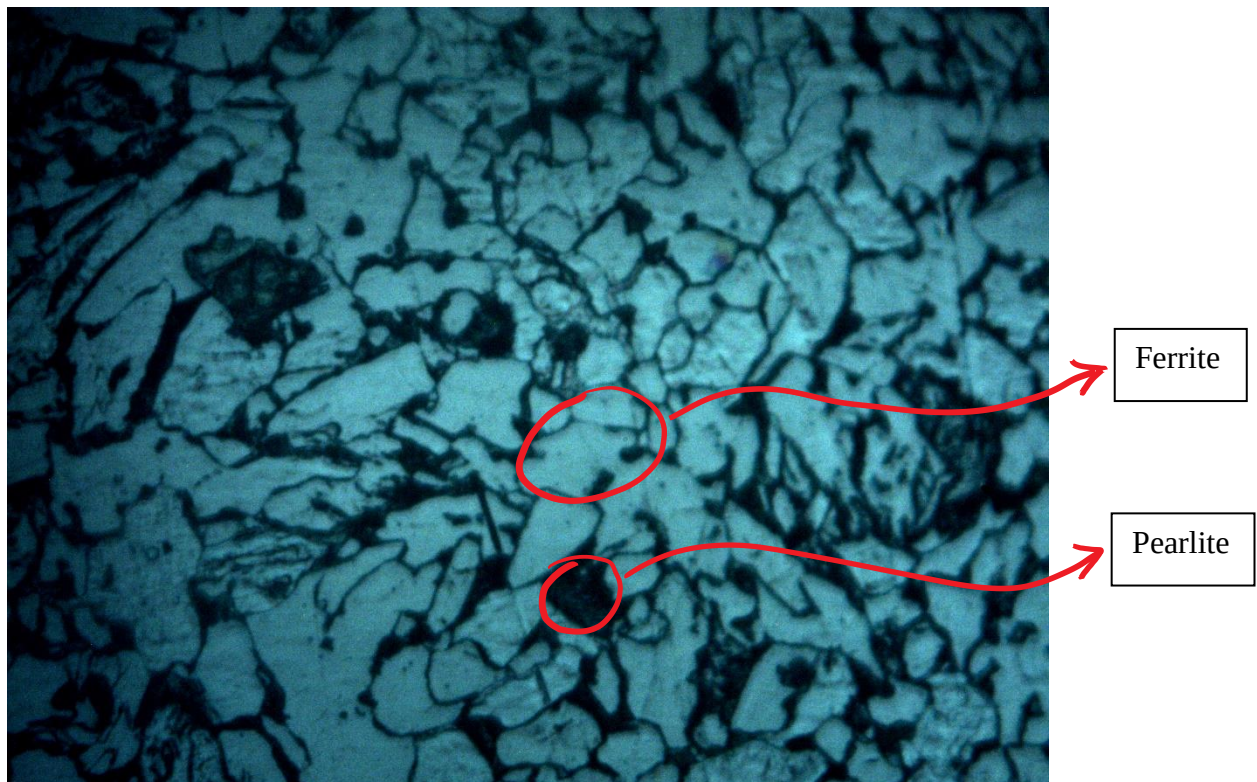


Figure 2: Impact test(-75°C)

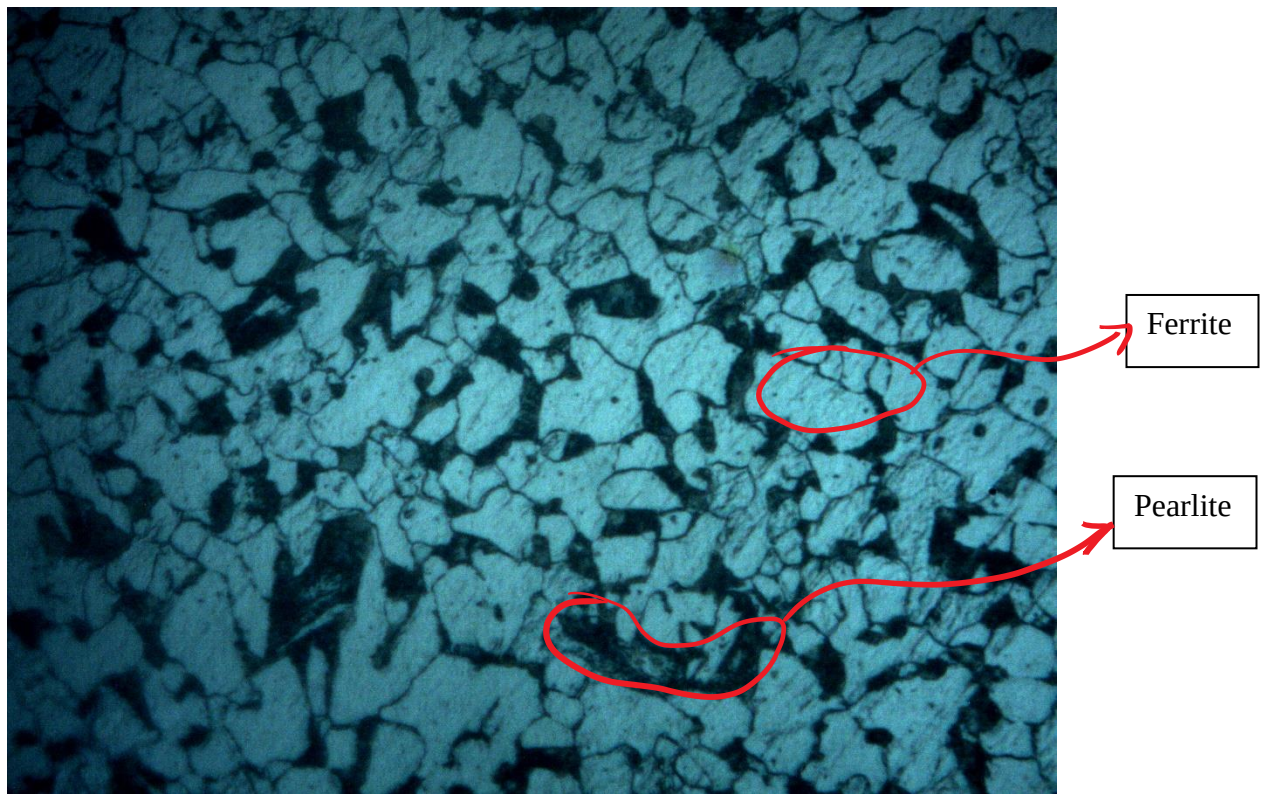


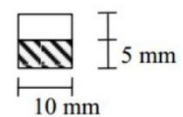
Figure 3: Impact test (14°C)

Calculations:

For -75°C,

Here impact energy=242J

$$\text{Area} = 10 \times 5 \text{ mm}^2 \\ = 50 \text{ mm}^2$$



Cross Section

$$\text{Impact toughness} = \frac{242 \text{ J}}{50 \text{ mm}^2} = 4.84 \text{ J/mm}^2$$

For -20°C,

Here impact energy=247J

$$\text{Impact toughness} = \frac{247 \text{ J}}{50 \text{ mm}^2} = 4.94 \text{ J/mm}^2$$

For 14°C,

Here impact energy=200J

$$\text{Impact toughness} = \frac{200}{50} \frac{J}{mm^2} = 4J/mm^2$$

For 100°C,

Here impact energy=225J

$$\text{Impact toughness} = \frac{225}{50} \frac{J}{mm^2} = 4.5J/mm^2$$

Results:

From the microstructure, I'm assuming that this mild steel has gone through a **normalizing** process. I'll show later in the discussion why I assuming this option strictly.

Data table from calculations:

Material type and test temperature(°C) (Mild steel)	Impact Energy(J)	Impact toughness (J/mm^2)
-75	242	4.84
-20	247	4.94
14	200	4
100	225	4.5

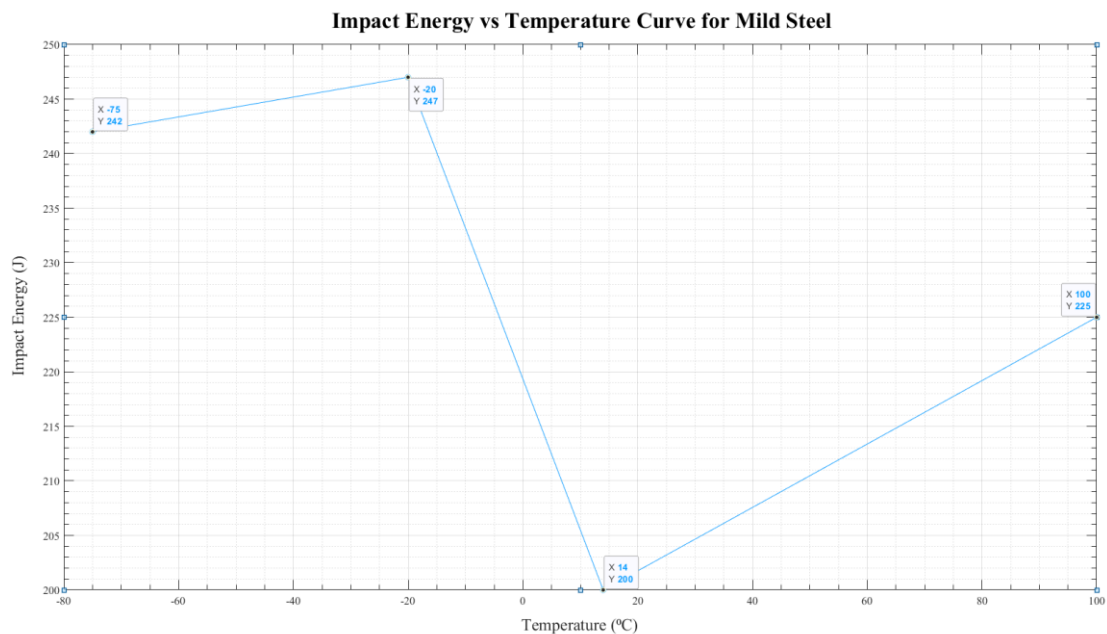


Figure 4: Impact test vs Temperature curve

Discussions:

1) From the impact test, we have come to know how a material's behavior will change with temperature variation. Impact loads are sudden and intense. Understanding how a material responds to such forces is crucial for applications such as tools, structures subjected to shocks or vibrations, and components in vehicles or machinery. Brittle materials tend to fracture with little energy absorption, while ductile materials can deform significantly before breaking. Impact tests help distinguish between these behaviors, guiding the selection of suitable materials for specific applications.

From our graph, energy absorption is high at low temperatures, indicating ductile behavior, and at high temperatures, it shows brittle behavior as energy absorption is low. So, this graph is crucial for selecting the appropriate material in different environments. Our mild steel performs well at low temperatures rather than high temperatures. What works at low temperatures, our material fits there best. (M.D. Bovea, 2006)

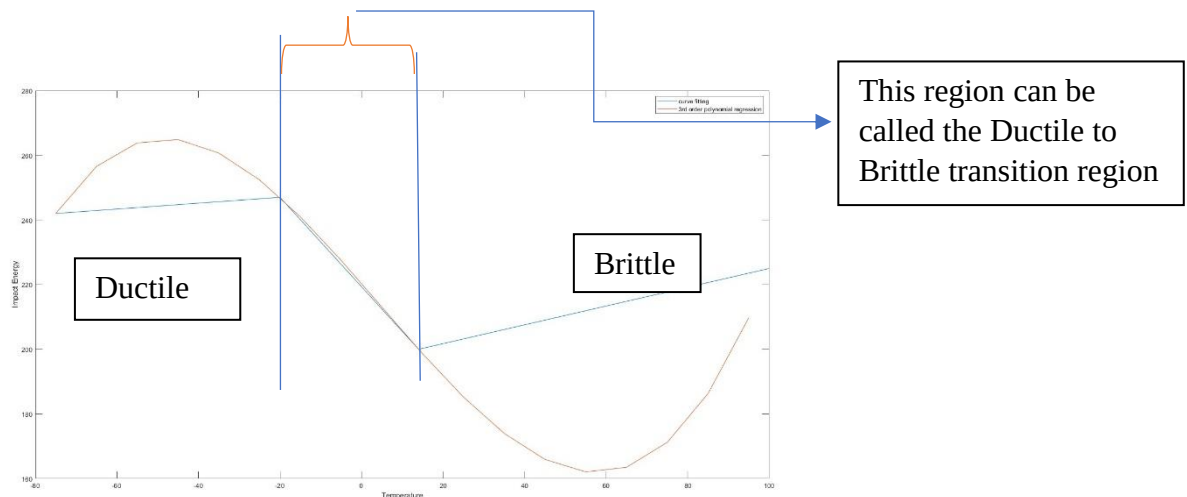


Figure 5: Curve using 3rd order polynomial regression

2) When it comes to choosing materials, the ductile-to-brittle transition phenomenon plays a significant role. It informs us about when the material will fail and whether it is suitable for our required environment. We can determine its safety and effectiveness under certain conditions.

In the ductile region, the material exhibits plastic deformation and deforms plastically in the presence of overload. It can show us when it is about to fracture by certain changes. On the other hand, brittle materials fail without any warnings, which can be catastrophic in crucial applications. The Ductile-to-Brittle Transition Temperature (DBTT) is the temperature at which a material transitions from ductile to brittle behavior. Below this temperature, the risk of brittle fracture increases dramatically.

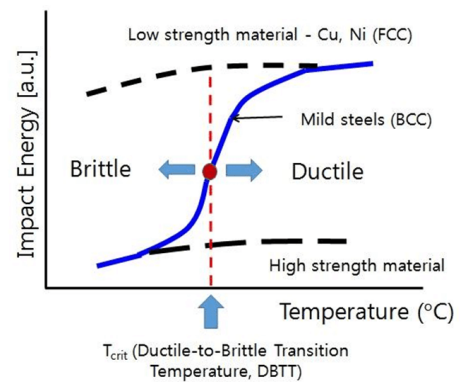
In our DBT curve, the ductile-to-brittle transition occurs between -20°C to 14°C . The microstructure shows some brittle and some ductile failure in the microstructure (impact test result). Below this region, the material is ductile, and above this region, it becomes brittle.

Applications:

Steel Bridges: Steels with high DBTTs are avoided in cold regions, while low-DBTT steels ensure safe operation even in harsh winters.

Plastic Car Bumpers: Polymers with DBTTs below freezing temperatures are chosen to resist cracking and ensure passenger safety in cold weather collisions.

Cryogenic Applications: Materials like stainless steel or special alloys with very low DBTTs are necessary for equipment operating at extremely low temperatures.
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3) The brittleness of some metals at low temperatures is a complex phenomenon influenced by several factors, but the main reason is to reduced atomic mobility:

- **Decreased thermal vibrations:** At lower temperatures, atoms in the metal's crystal lattice vibrate less, making them "stickier" and less prone to sliding past each other. This impedes the movement of dislocations, which are tiny line defects that allow metals to deform plastically (bend without breaking). (Riley, 2019)
- **Grain boundary effects:** Grain boundaries are the junctions between microscopic crystals within a metal. They can act as crack initiation points, and at low temperatures, the reduced atomic mobility makes cracks more likely to propagate along these boundaries, leading to brittle fracture.
- **Interstitial impurities:** Some impurities, particularly small atoms that can fit into the spaces between metal atoms (called interstitial impurities), can hinder dislocation movement and further reduce the material's toughness at low temperatures.

Beyond these main factors, other contributing elements to low-temperature brittleness include:

Microstructure: The size and shape of the metal's grains can affect its brittleness at low temperatures.

Chemical composition: The specific elements present in the metal can influence its susceptibility to brittleness.

Processing history: The way the metal was processed and heat-treated can affect its internal structure and, consequently, its brittleness at low temperatures. (I. Milne, 2003)

4) In our sample, we observed ductile fracture, as none of our samples tore apart. After the impact test, all our samples exhibited a similar appearance. Therefore, we can conclude that the temperature was not sufficiently low to cause the two pieces to separate from each other..

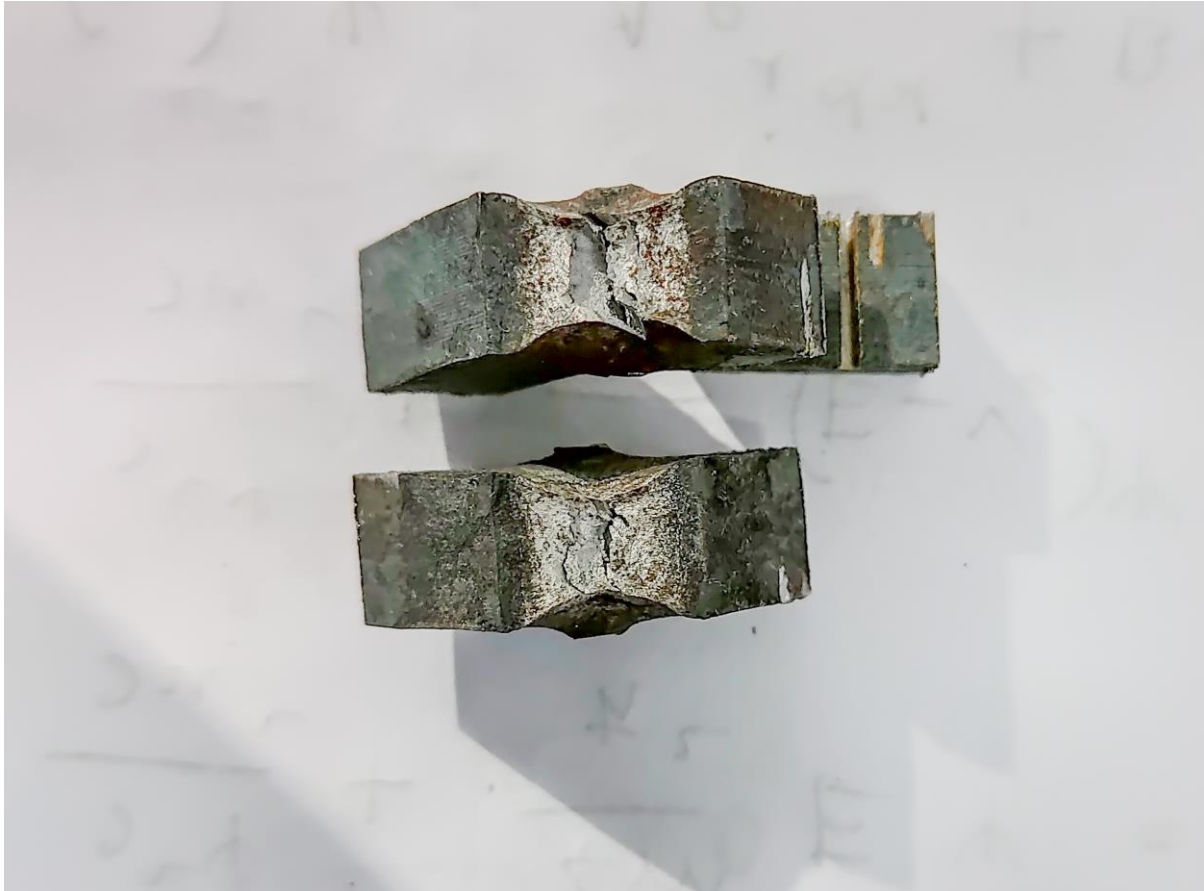
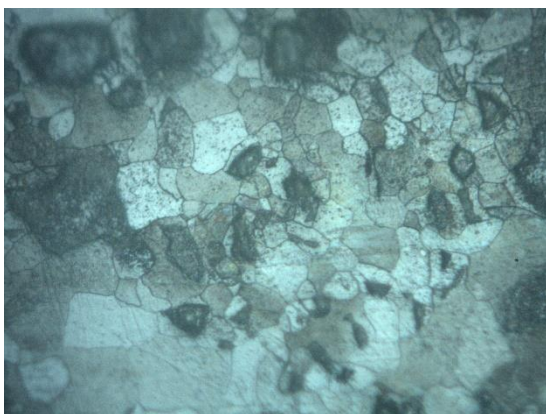
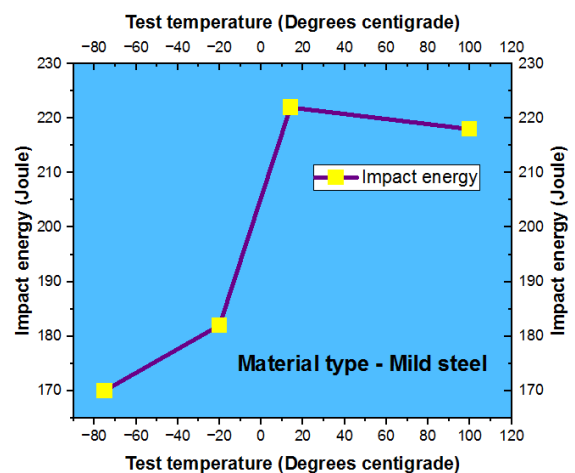


Figure 6: Samples after impact test.

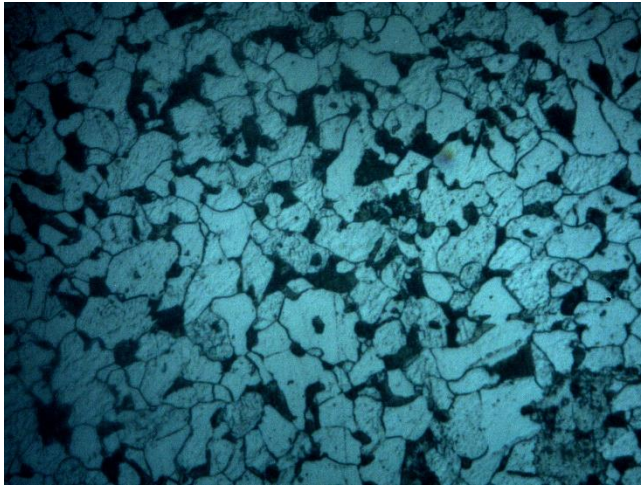
Microstructures and curve of group-01



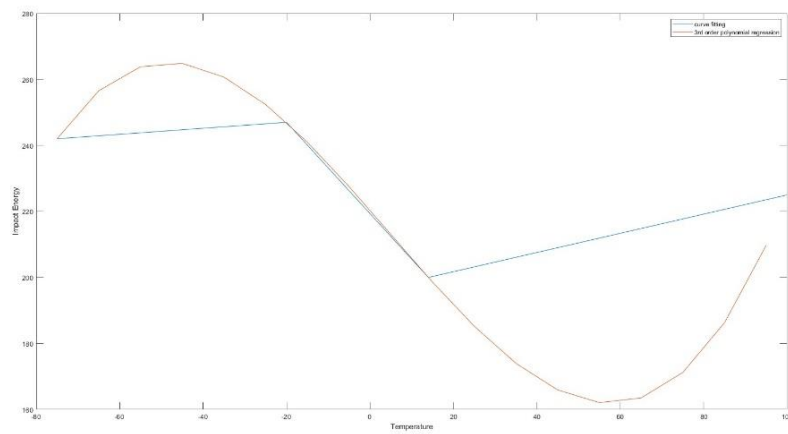
Heat treatment: Normalizing



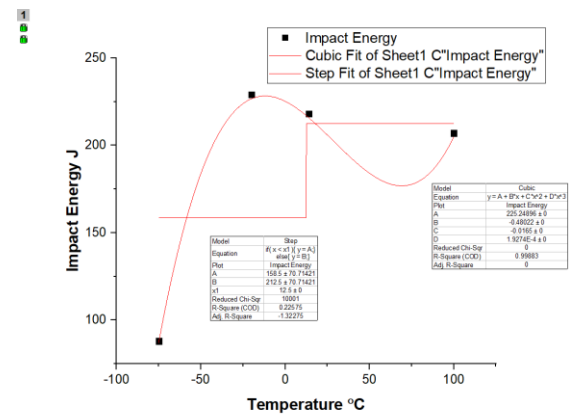
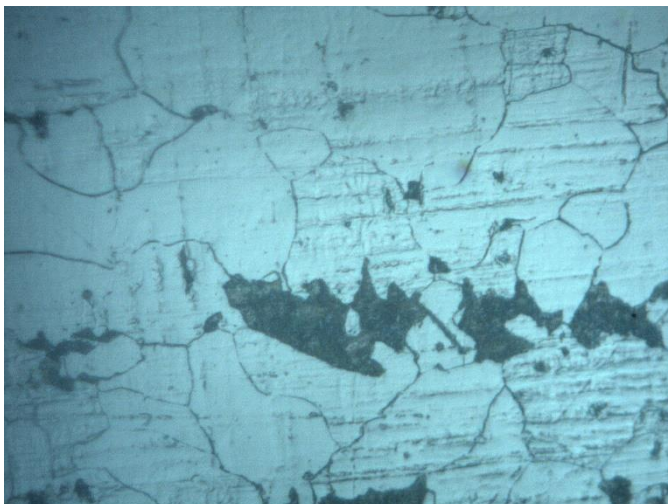
Microstructure and curve of group-02



Heat treatment: Normalizing

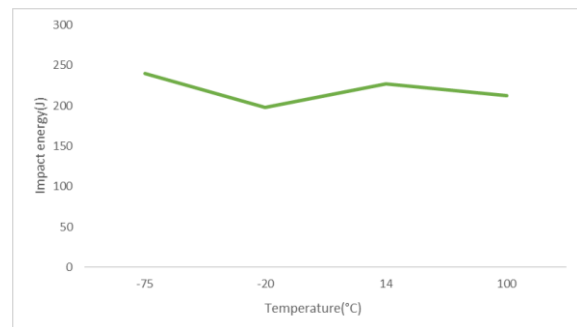
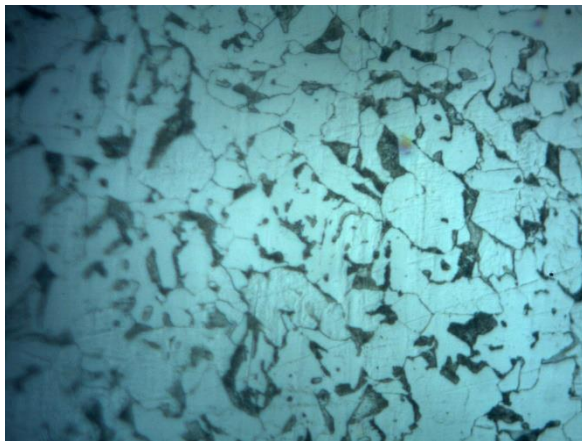


Microstructure and curve of group-03



Heat treatment: Annealing

Microstructure and curve of group-04



We (Group-02) obtained a curve that is completely different from everyone else's. When compared with the original DBT curve, our curve is entirely reversed. We assume that the third value of our impact test, which is 200J at room temperature (14°C), was recorded incorrectly. If our impact test value was more than 247J at 14°C, our graph would fit more correctly and support the theory.

It is challenging to determine the heat treatment process our sample has undergone by analyzing its microstructure. It could be a normal heating process without a complex heat treatment procedure, or it could be a normalizing process it has gone through. This is suggested by the dense and fine grains in our microstructure. An example of a normalized structure:

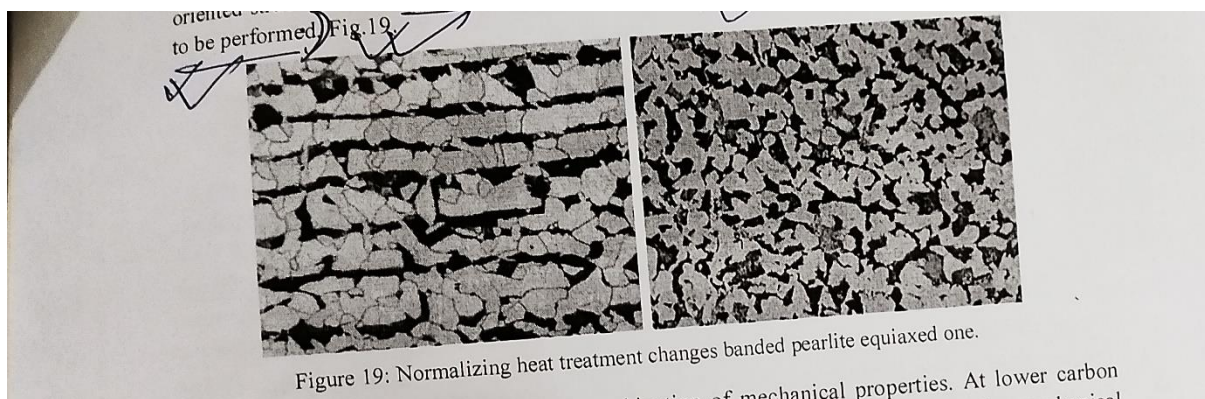


Figure 7: Proper normalizing heat treatment microstructure.

Another possibility could be quenching. Water quenching in a fast manner means specimens were dipped into water and removed within a few seconds, which led to the formation of new grains but did not allow enough time for proper development. As a result, some grains were obstructed by each other, indicating that they are not properly connected through grain boundaries.

The curve of Group-01 and Group-03 fits properly, but for Group-04, their DBTT curve fluctuates. My assumptions are:

- 1) High carbon content in specimens.
- 2) Slight possibility of having a wrong impact test result.
- 3) The heat treatment process was not done properly.

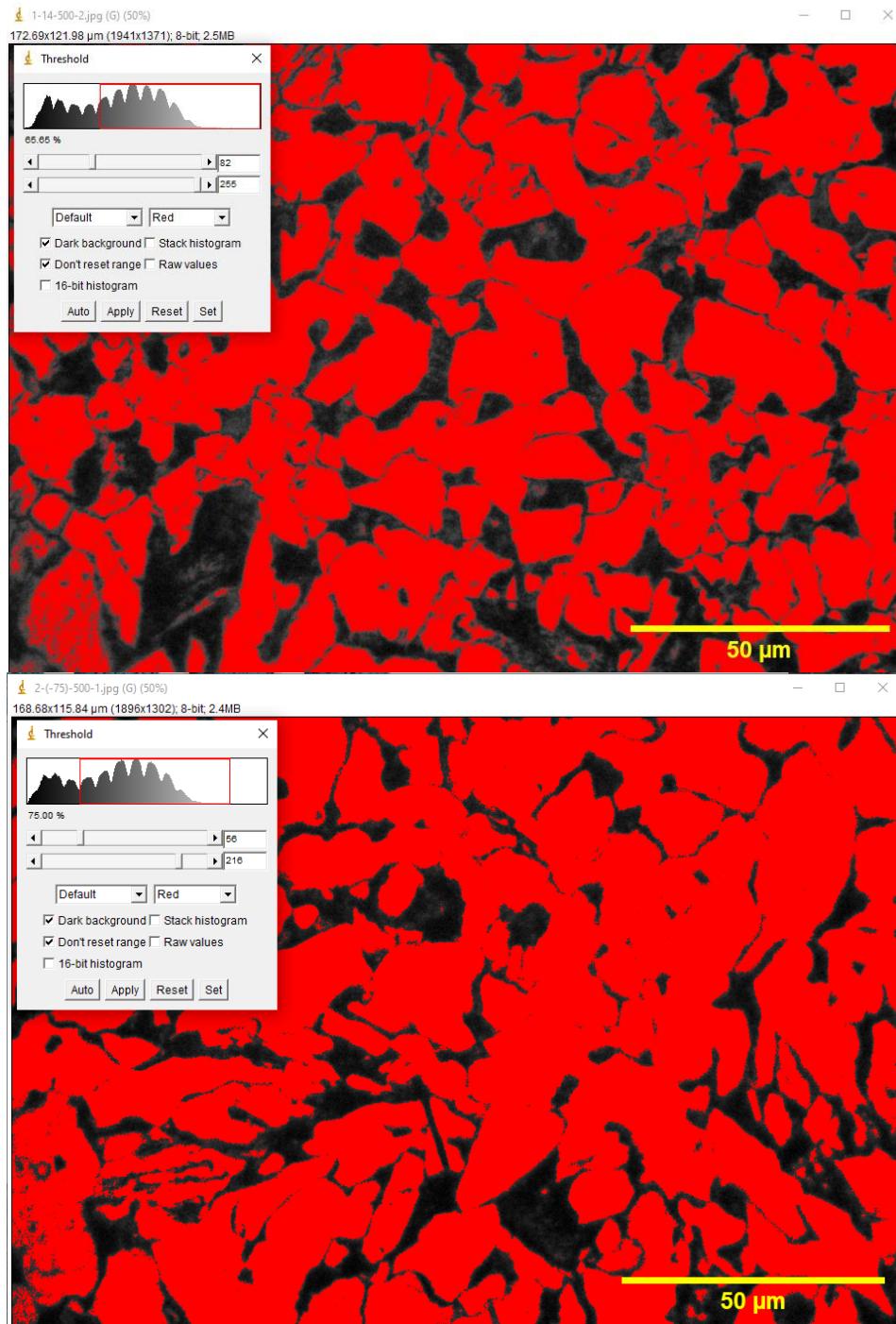


Figure 8: Analyses using imagej software, to find out ferrite and pearlite percentage

By analyzing the picture we can see, ferrite %=66.65 and pearlite %=33.35 at 14°C & ferrite=75% and pearlite=25% at -75°C.

Here, when temperature increases, ferrite percentage decreases. To explain this, we have to know about the structure of ferrite, which is BCC. BCC has less carbon than FCC. When we increase temperature extra carbon comes out because the solubility of carbon in BCC decreases. Another factor could be, that alloying elements such as Mn, Cr, Ni, etc. can destabilize ferrite and let carbon to come out to form a pearlitic zone. (L. I. Gladshtein, 2012)

References

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